

Convenções:

Here $\lambda = 1/T$.

The notation 0 means lifelong (or probably lifelong) immunity, hence $T = \infty$ and $\lambda = 0$.

N/D means “not determined”: the available official evidence does not give a sufficiently well-defined characteristic duration.

N/C means “not comparable”, in particular when the ratio is 0/0.

Characteristic duration of natural immunity is greater than that of vaccine-induced immunity.

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Table 1: $\psi < \theta$. Imunidade natural maior que a vacinal.

	T_{natural}	T_{vaccine}	λ_{natural}	λ_{vaccine}	$\lambda_{\text{natural}}/\lambda_{\text{vaccine}}$
Sarampo	permanente	Possivelmente permanente	0	0	N/C
Rubéola	Essencialmente permanente	Pelo menos 15 anos	0	$\lesssim 1.83 \times 10^{-4}$	0
Varicela	duração longa	Possivelmente permanente	0	0	N/C
Hepatite A	Essencialmente permanente	$\geq 20\text{-}25$ yr	0	$\leq 1.37 \times 10^{-4}$	0

Table 3: $\psi > \theta$. Imunidade vacinal maior que a natural.

Disease	T_{natural}	T_{vaccine}	λ_{natural}	λ_{vaccine}	$\lambda_{\text{natural}}/\lambda_{\text{vaccine}}$
Coqueluche	4-20 yr	4-5 yr	1.71×10^{-4} – 6.85×10^{-4}	5.48×10^{-4} – 6.85×10^{-4}	≈ 0.25 – 1.25
Influenza	$\sim$ one season	$\sim$ one season	$\sim 2.74 \times 10^{-3}$	$\sim 2.74 \times 10^{-3}$	≈ 1
HPV	Sem proteção natural	≥ 10 – 12 yr	N/D	$\lesssim 2.74 \times 10^{-4}$	N/D
COVID-19	~ 6 – 8 meses	longer for severe	$\sim 4.1 \times 10^{-3}$ – 5.5×10^{-3}	$\sim 4.1 \times 10^{-3}$ – 5.5×10^{-3}	≈ 1